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9.3.4 The Rayleigh quotient Iteration

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know that lambda and x are an eigenvalue and eigenvector of A, can I somehow use that to come up with convenient eigenvalues and eigenvectors of the matrix A minus rho I quantity inverse? Okay? This is known as a shifted matrix. You're shifting A by rho.

But we're interested in here what happens to the matrix when we invert it.

And you're also going to look at the question of if I know X and Lambda such

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Reading assignment

1/1 point (graded)

Read Unit 9.3.4 of the notes. [\[LINK\]](#)

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Homework 9.3.4.1

1/1 point (graded)
Let $A \in \mathbb{C}^{m \times m}$, $\rho \in \mathbb{C}$, and (λ, x) an eigenpair of A .

Which of the follow is an eigenpair of the shifted matrix $A - \rho I$:

- ☐ (λ, x)
- ☐ $(1/\lambda, A^{-1}x)$
- ☒ $(\lambda - \rho, x)$
- ☐ $(1/(\lambda - \rho), x)$



Solution

For a complete solution, see the [notes](#).

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Answers are displayed within the problem

Homework 9.3.4.2

1/1 point (graded)
Let $A \in \mathbb{C}^{m \times m}$, $A = X\Lambda X^{-1}$ and $\rho \in \mathbb{C}$. Prove that $A - \rho I = X(\Lambda - \rho I)X^{-1}$.

- ☒ done



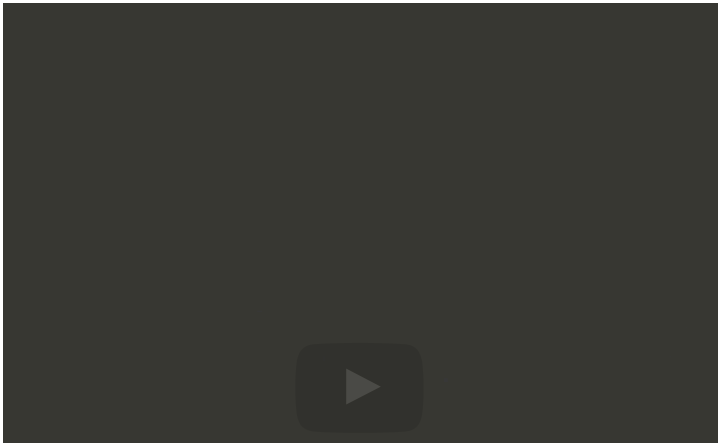
Solution

For a complete solution, see the [notes](#).

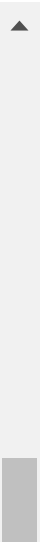
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large relative to this next ratio. Okay? So what does this suggest? This suggests that if we have a good idea of what the smallest eigenvalue is associated with a matrix, then we can instead iterate with the shifted matrix





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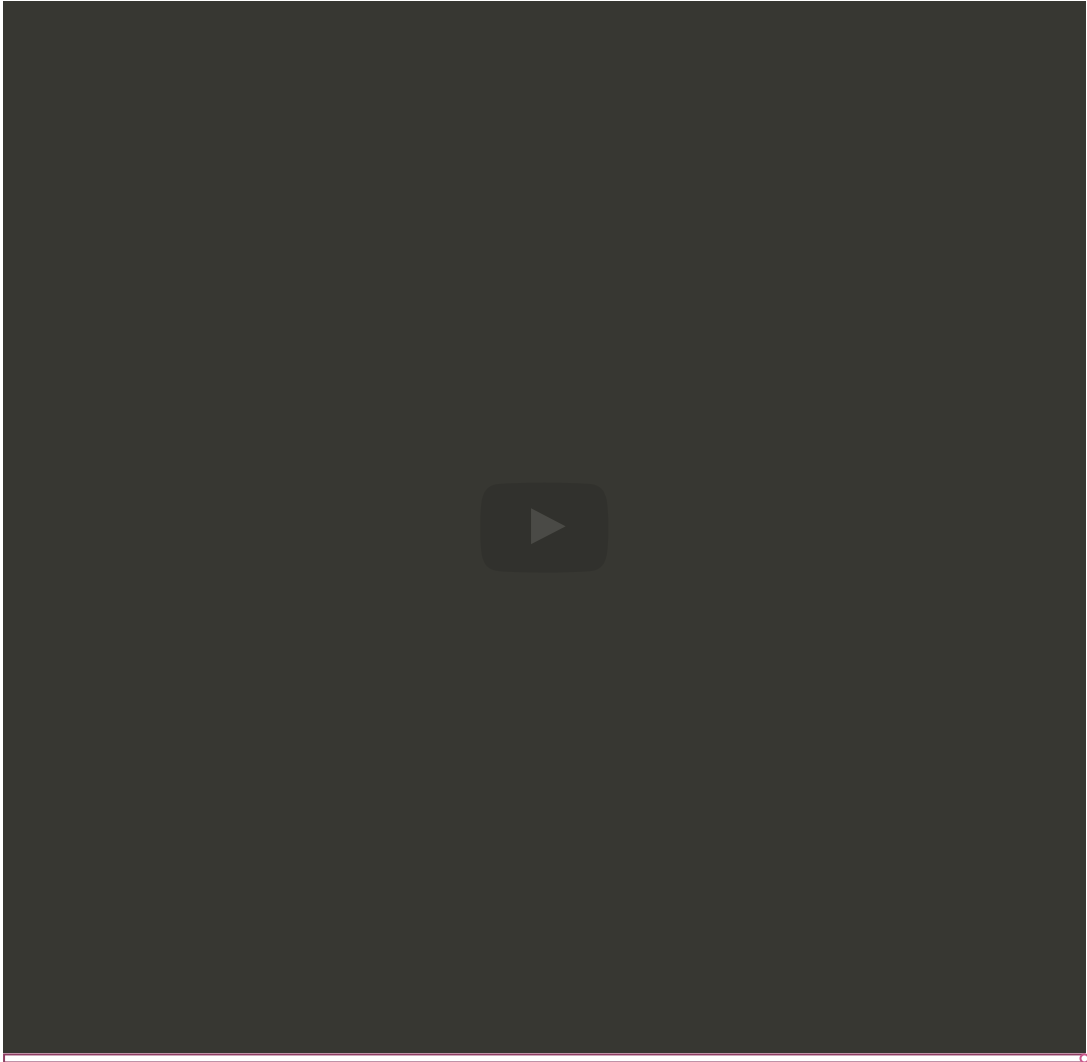
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inverse. Okay we obviously in our analysis then would want to multiply by this right here. In practice we would do what we always do, which is keep our vector of length 1 in every step by normalizing it. What



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